③ let $u = e^{cx}$, $dv = \sin x\,dx$

$\therefore du = ce^{cx}dx$, $v = -\cos x$

$\therefore I = -c\int_0^{\pi} u\,dv$

$= -c\left(uv\Big|_0^{\pi} - \int_0^{\pi} v\,du\right)$

$= -c\left(e^{cx}(-\cos x)\Big|_0^{\pi} - \int_0^{\pi} -\cos x\, ce^{cx}dx\right)$

$= -c\left[(e^{c\pi}+1) + \int_0^{\pi} ce^{cx}\cos x\,dx\right]$

$= -c\left(e^{c\pi}+1+cI\right)$

④ $\therefore I = I_0 = \dfrac{e^{c\pi}+1}{(-\frac{1}{c}-c)}$

$= -\dfrac{ce^{c\pi}+c}{c^2+1}$

2.3 test answer: check the special cases to see if we've made a mistake

① $I = \int_0^{\pi} e^{cx}\cos x\,dx < 0$ if $c>0$ and $I_0 < 0$, if $c>0$.

② $I = \int_0^{\pi}\cos x\,dx = 0$ if $c=0$ and $I_0 = 0$, if $c=0$.

② method 2.

let $x_B = P(\text{Bird wins})$

$\because$ memoryless process

$\therefore x_B = p + rx_B$

$\therefore x_B = \dfrac{p}{1-r}$

e.g., if $p=q=\frac{1}{2} \Rightarrow x_B = \frac{2}{3}$

2. Bring It Over Method

$I = \int_0^{\pi} e^{cx}\cos x\,dx$

2.2 solve:

① $\int_0^{\pi} u\,dv + \int_0^{\pi} v\,du = uv\Big|_0^{\pi}$

② let $u = e^{cx}$, $dv = \cos x\,dx$

$\therefore du = ce^{cx}dx$, $v = \sin x$

$\therefore I = \int_0^{\pi} u\,dv = uv\Big|_0^{\pi} - \int_0^{\pi} v\,du$

$= e^{cx}\sin x\Big|_0^{\pi} - \int_0^{\pi}\sin x\, ce^{cx}dx$

$= -c\int_0^{\pi} e^{cx}\sin x\,dx$

Basketball Problem

1. let p be the chance Bird has of making a basket, q the chance Magic has. If Bird shoots first, what's the pr that he wins the shootout?

solve:

① method 1

$$\begin{cases} p \\ 1-p \begin{cases} q \\ 1-q \begin{cases} p \\ 1-p \begin{cases} q \\ 1-q \begin{cases} p \\ 1-p \end{cases} \end{cases} \end{cases} \end{cases} \end{cases}$$

$\therefore Pr(\text{Bird wins}) = p + rp + r^2p + \cdots$

$= \sum_{n=0}^{\infty} r^n p = \dfrac{p}{1-r}$,

here $r = (1-p)(1-q)$

5. Alice, Bob and Charlie are playing game of cards. The first one to draw a diamond wins. They take turns drawing. Alice the Bob then Charlie then Alice and so on. If card isn't diamond it's put back in the deck and deck is shuffled. What's the pr that each person wins?

solve: ① method 1.

let $x = P(\text{Alice wins})$, $y = x_B = Pr(\text{Bob wins})$,

$z = x_C = Pr(\text{Charlie wins})$.

$\therefore x = \frac{1}{4} + \frac{3}{4}\frac{3}{4}\frac{3}{4}x$,

$y = \frac{3}{4}\frac{1}{4} + \frac{3}{4}\frac{3}{4}\frac{3}{4}y$

$z = \frac{3}{4}\frac{3}{4}\frac{1}{4} + \frac{3}{4}\frac{3}{4}\frac{3}{4}z$

$\therefore x = \frac{16}{37}$, $y = \frac{12}{37}$, $z = \frac{9}{37}$

(test answer: $x+y+z=1$.)

② method 2: $\because x_B = \frac{3}{4}x_A$, $x_C = \frac{3}{4}\frac{3}{4}x_A = \frac{3}{4}x_B$

$\because x_A + x_B + x_C = 1$

$\therefore (1 + \frac{3}{4} + \frac{3}{4}\frac{3}{4})x_A = 1$

$\Rightarrow x_A = \frac{16}{37}$, $x_B = \frac{3}{4}x_A = \frac{12}{37}$, $x_C = \frac{9}{37}$

4. $F_{n+2} = F_{n+1} + F_n$, $F_0 = 0$, $F_1 = 1$

$\Rightarrow x = \sum_{n=0}^{\infty} \frac{F_n}{3^n}$

solve:

$\because x = \sum_{n=0}^{\infty} \frac{F_n}{3^n}$

$= \frac{F_0}{1} + \frac{F_1}{3} + \sum_{n=2}^{\infty} \frac{F_n}{3^n}$

$= \frac{1}{3} + \sum_{m=0}^{\infty} \frac{F_{m+2}}{3^{m+2}}$

$= \frac{1}{3} + \sum_{m=0}^{\infty} \frac{F_{m+1} + F_m}{3^{m+2}}$

$= \frac{1}{3} + \frac{1}{3}\sum_{m=0}^{\infty} \frac{F_{m+1}}{3^{m+1}} + \frac{1}{9}\sum_{m=0}^{\infty} \frac{F_m}{3^m}$

$= \frac{1}{3} + \frac{1}{3}x + \frac{1}{9}x$

(note $\sum_{m=0}^{\infty} \frac{F_{m+1}}{3^{m+1}} = \sum_{n=1}^{\infty} \frac{F_n}{3^n}$

$= \sum_{n=0}^{\infty} \frac{F_n}{3^n}$)

$\therefore x = \frac{3}{5}$

3. $\sum_{n=0}^{\infty} s^n = \frac{1}{1-s} \Rightarrow \sum_{n=0}^{\infty} (-s)^n = \frac{1}{1-(-s)}$

$s \in (0,1)$

proof:

$\because \sum_{n=0}^{\infty} (-s)^n = (-s)^0 + (-s)^1 + (-s)^2 + \cdots$

$= (-s)^0 + (-s)^2 + (-s)^4 + \cdots$

$+ (-s)^1 + (-s)^3 + (-s)^5 + \cdots$

$= \sum_{n=0}^{\infty} (-s)^{2n} + \sum_{n=0}^{\infty} (-s)^{2n+1}$

$= \sum_{n=0}^{\infty} (s^{2n} - s^{2n+1})$

$= (1-s)\sum_{n=0}^{\infty} s^{2n}$

$\therefore \sum_{n=0}^{\infty} (-s)^n = (1-s)\sum_{n=0}^{\infty} s^{2n}$

$= (1-s)\frac{1}{1-s^2}$

$= \frac{1}{1-(-s)}$